PRIVATE - CONTAINS A FULL SUPPLIED BSN - DO NOT PUBLISH

The Tom Klootwijk Manifold

A formally valid 7-dimensional compact manifold

$$\boxed{\mathcal{K}_{\text{TK}}(\mathbf{r}) = \prod_{i=0}^{6} S^1(r_i)}$$

$$\boxed{\mathcal{K}_{\text{TK}}(\mathbf{r}) = \left\{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2,\ i = 0, \ldots, 6\right\}}$$

Author	Tom Klootwijk
BSN supplied*	NL200678942
Date of birth supplied	10-07-1990
Age at occasion	36
Edition	Unity 6.3 / age 36 (U6.3-A36)
Occasion local time	2026-08-22T06:34:39+02:00 (Europe/Amsterdam request capture)
Occasion UTC	2026-08-22T04:34:39Z
Document ID	TKM-U63-A36-20260822-063439
Author-record SHA-256	`7d85cc5019309659b8279866b0bf6dde69ef18373d25dc71051fe8824ee2e3b4`
Target	Unity 6000.3.22f1; URP 17.3.0

The supplied BSN string is reproduced only because the author expressly requested it. Its nine-digit numeric portion passes the arithmetic 11-test, but this does not verify identity, issuance, ownership, priority, or legal authorship.

Contents

1 Verdict

The source dossier contains a useful creative direction but not a valid theorem as written. Its strongest defensible ingredients are: seven dimensions for seven orthogonal directions; compact procedural notation; rounded rather than jagged visual factors; deterministic personalization; and a GPU-friendly visualization. Its claims that 0-indexing changes Hamming distance, that a parity bit can perform the 943-to-937 transition, that SDF $= 0$ automatically creates a sphere, and that the rendering has zero VRAM or constant frame cost are false.

This specification replaces those claims with a standard, self-contained object: a product of seven circles. The result is a smooth, compact, connected, intrinsically flat 7-manifold with a global orthonormal frame. The chosen personal name is a nomenclature decision for this document; it is not a claim of external peer-review recognition or legal registration.

One-line definition

$$\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) := \prod_{i=0}^{6} S^1(r_i), \qquad r_i > 0.$$

2 Most compact literal definitions

The following are equivalent descriptions of the same manifold.

2.1 Product literal

$$\boxed{\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) = S^1(r_0) \times \cdots \times S^1(r_6)}.$$

2.2 0-indexed implicit literal

$$\boxed{F_i(x) = x_{2i}^2 + x_{2i+1}^2 - r_i^2 = 0, \quad i = 0, \ldots, 6}.$$

Thus $\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) = F^{-1}(0) \subset \mathbb{R}^{14}$, with $F : \mathbb{R}^{14} \to \mathbb{R}^7$.

2.3 Parametric literal

For $\theta \in (\mathbb{R}/2\pi\mathbb{Z})^7$ and author phase $\phi(\alpha)$,

$$\boxed{\Phi_\alpha(\theta)_{2i} = r_i \cos(\theta_i + \phi_i), \quad \Phi_\alpha(\theta)_{2i+1} = r_i \sin(\theta_i + \phi_i)}.$$

The phase changes only the chosen coordinates/visual instance; it does not alter the manifold's diffeomorphism type.

2.4 Metric and orthogonality literal

$$\boxed{g = \sum_{i=0}^{6} r_i^2\, d\theta_i^2, \qquad E_i = r_i^{-1}\partial_{\theta_i}, \qquad g(E_i, E_j) = \delta_{ij}}.$$

This is the exact formal statement that there are seven pairwise perpendicular unit tangent directions at every point.

2.5 Code literal

```
for (int i = 0; i < 7; ++i) {
    x[2*i]   = r[i] * cos(theta[i] + phase[i]);
    x[2*i+1] = r[i] * sin(theta[i] + phase[i]);
}
```

3 Proof of validity

3.1 Smoothness and dimension

Define $F_i(x) = x_{2i}^2 + x_{2i+1}^2 - r_i^2$. At any $x \in F^{-1}(0)$, the i-th row of DF_x is zero except in the pair $(2i, 2i+1)$, where it equals $(2x_{2i}, 2x_{2i+1})$. Because $r_i > 0$, this pair is nonzero. The seven rows have disjoint supports, hence are linearly independent. Therefore rank $DF_x = 7$ everywhere on the zero set. By the regular-value theorem, $F^{-1}(0)$ is a smooth embedded manifold of dimension $14 - 7 = 7$.

3.2 Global orthonormal frame

The tangent vector $\partial\Phi/\partial\theta_i$ is nonzero only in coordinate pair i:

$$\frac{\partial\Phi}{\partial\theta_i} = (0, \ldots, -r_i \sin(\theta_i + \phi_i), r_i \cos(\theta_i + \phi_i), \ldots, 0).$$

Distinct tangent vectors have disjoint supports and dot product zero; the squared norm of the i-th vector is r_i^2. Normalizing gives the seven global orthonormal fields E_i.

3.3 Compactness, topology, and roundness

Each circle is compact and connected, so their finite product is compact and connected. Every factor is round, but the product is a 7-torus, not a sphere. It is intrinsically flat under the product metric even though its embedding in $\mathbb{R}^{14}$ is extrinsically curved. This distinction supplies a precise version of "round without jaggedness" without pretending that SDF $= 0$ creates a hypersphere.

4 What was and was not validated from the source

Claim	Verdict	Reason
Seven perpendicular directions	Valid in dimension 7	The frame $E_0, \ldots, E_6$ is orthonormal.
Base-4 values	Valid arithmetic	$973 = 33031_4$, $943 = 32233_4$, $937 = 32221_4$.
943 to 937 by one bit	Invalid	$943 \oplus 937 = 6 = 110_2$, so the Hamming distance is 2.
0-index/LSB changes that distance	Invalid	Relabeling bit positions does not change XOR or Hamming distance.
Parity bit executes the mutation	Invalid	Parity is a check value, not a general state-transition mechanism.
SDF $= 0$ makes a sphere	Invalid	Zero specifies a level set; geometry comes from the field being evaluated.
Procedural storage is compact	Conditionally valid	Parameters can be compact, but evaluation still costs work and memory.
Zero VRAM / $O(1)$ frame cost	Invalid	Render targets, code/resources, samples, vertices, and pixels remain.

5 Personalization without corrupting the mathematics

Let α be the SHA-256 fingerprint of the private canonical author record. This specification uses

$$\alpha = \texttt{7d85cc5019309659b8279866b0bf6dde69ef18373d25dc71051fe8824ee2e3b4}.$$

Seven phases and seven 3D projection planes are deterministically derived from α. The private supplied BSN string is not used as a mathematical premise, and it is not embedded in runtime code. This keeps authorship metadata separate from proof.

Index	Phase (radians)	Slice frequency (m_i, n_i)	Angular speed
0	3.080712791	(1, 0)	0.17
1	5.014583196	(0, 1)	-0.13
2	0.618194258	(1, 1)	0.11
3	3.690086659	(1, -1)	-0.19
4	4.519778518	(2, 1)	0.07
5	3.740420404	(1, 2)	0.23
6	4.338001794	(2, -1)	-0.09

6 Unity 6.3 implementation

6.1 Exact implementation layer

`TKManifoldMath` implements the 14-coordinate embedding, seven implicit residuals, and tangent vectors. Runtime tests verify constraint error and the tangent Gram matrix.

6.2 Renderable 3D layer

A screen cannot directly display a 7D manifold. The mesh component evaluates

$$X(u, v, t) = P\big(\Phi_\alpha(A_t(u, v))\big),$$

where $A_t : T^2 \to T^7$ is the integer-frequency phase slice $\theta_i = m_i u + n_i v + \omega_i t + \phi_i$, and $P : \mathbb{R}^{14} \to \mathbb{R}^3$ is a fixed linear projection. The renderer therefore shows a 3D projection of a 2D slice. Self-intersections are allowed, and seven-way orthogonality is not claimed after projection.

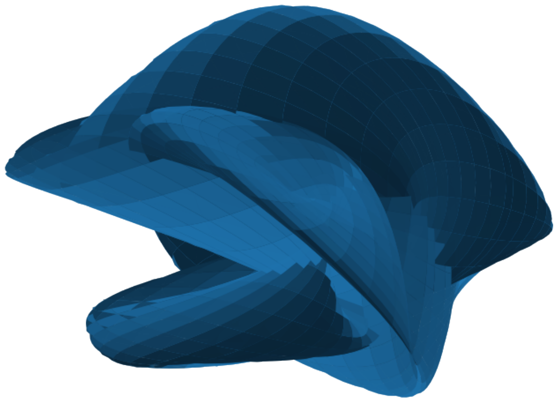

Figure 1: A generated 3D projection of the personalized periodic 2D slice. This is a visualization, not the whole 7D manifold.

6.3 Optional SDF surrogate

The package also includes a rounded 3D field made from seven capsule distance functions and a smooth-min union. It is useful as a procedural art object but is explicitly labeled a surrogate. Its per-pixel work is bounded by roughly seven distance evaluations per march step, not constant time.

6.4 Default complexity

For $N_u = 96$ and $N_v = 48$, the mesh has $(N_u + 1)(N_v + 1) = 4{,}753$ vertices and $2N_uN_v = 9{,}216$ triangles. Animated CPU deformation is $O(7N_uN_v)$ per update and mesh memory is $O(N_uN_v)$. The SDF surrogate is $O(7S)$ field operations per shaded pixel for at most S march steps, plus normal estimation and framebuffer memory.

6.5 Installation

1. Open a Unity 6000.3 URP project; the delivery targets editor 6000.3.22f1 and URP 17.3.0.
2. In Package Manager choose **Add package from disk**. Select `package.json` from the delivered package folder `com.tomklootwijk.manifold`.
3. Use **GameObject** > **Tom Klootwijk Manifold** and choose **Create Projected 7-Torus Slice**.
4. Optionally create the rounded SDF surrogate; keep its cube scale uniform.
5. Run W**indow** > **General** > **Test Runner** and execute the package tests.

7 Validation record

The independent Python validator reproduced the author fingerprint and checked the base-4 values, XOR result, Hamming distance, 250 random manifold samples, seven-circle residuals, tangent orthogonality and norms, and the personalized projection planes. Numerical errors were at floating-point roundoff scale. Here, formal validation means an explicit definition and complete mathematical proof plus executable checks; the proof was not machine-checked in a theorem prover.

An actual Unity Editor import, C# compilation, shader compilation, and GPU run were **not** executed in the build environment because the Unity Editor was unavailable. The archive includes NUnit runtime tests so the remaining validation occurs immediately in the target editor rather than being silently asserted here.

8 Nomenclature and privacy status

"The Tom Klootwijk Manifold" is the author's selected label for the object defined in this document. The full supplied BSN string appears only in the private copy and private author record. The public runtime package uses only the fingerprint. Keep this private copy out of public repositories, issue trackers, screenshots, and asset-store submissions.

9 References and implementation basis

- Supplied source dossier, especially its proposed name/notation, its 7D discussion, and its internal critique of geometry, topology, and information theory.
- Unity 6.3 User Manual and Unity 6000.3.22f1 release documentation.
- Universal Render Pipeline 17.3.0 package documentation; the package manifest pins this version for the Unity 6000.3 target.
- Standard regular-value theorem and product-manifold construction; the proof needed here is given in full above.

A Literal definition card

$$\begin{aligned}
\mathcal{K}_{\mathrm{TK}}(\mathbf{r}) &= \prod_{i=0}^{6} S^1(r_i) \\
&= \{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2,\ i = 0, \dots, 6\}, \\
\Phi_\alpha(\theta)_{2i:2i+1} &= r_i\big(\cos(\theta_i + \phi_i), \sin(\theta_i + \phi_i)\big), \\
g &= \sum_{i=0}^{6} r_i^2 d\theta_i^2, \qquad g(E_i, E_j) = \delta_{ij}, \\
X(u, v, t) &= P\Phi_\alpha\big((m_i u + n_i v + \omega_i t)_{i=0}^{6}\big).
\end{aligned}$$

B Private/public author-record note

The supplied BSN string is reproduced only because the author expressly requested it. Its nine-digit numeric portion passes the arithmetic 11-test, but this does not verify identity, issuance, ownership, priority, or legal authorship.